Sung Ha Hwang

Software Engineer, ex-Meta (Feed and Threads ranking) | Full-stack apps live on Google Play and the App Store
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WORK EXPERIENCE

Meta Jul 2024 – Jul 2026
Software Engineer, Modern Recommendation Systems (Facebook, Instagram, Threads)

- Integrated and launched a Llama-based quality model in Facebook Feed ranking: ∼380K incremental daily users (+0.0183% global DAU, +0.0444% US/CA) in a two-week A/B test.
- Saved 77 A100 hosts ($2.86M/year) by enabling off-peak backfill for a production ranking model.
- Built and productionized a KNN-based clustering algorithm for Threads recommendations, lifting time spent by 0.32%.
- Designed and launched an observability tool tracking reliability of 200+ ML features across ranking and retrieval; exposing live inference values cut debugging time by ∼40% and grew to 15K views/month.
- Won "Code of the Week" among 1,000+ engineers in a company-wide core reliability competition.

Northern Labs Sep 2022 – Dec 2022
Software Engineer Intern

- Built Elixir/TypeScript payment services on Kubernetes with SQL storage and GitHub CI/CD; added end-to-end tests and Slack-webhook alerting for early defect detection, improving system stability by 28%.

CGI Group Inc. Sep 2020 – Apr 2021
Java Developer Co-op

- Cut page load time by 50% for 10K+ daily users by parallelizing data retrieval with asynchronous Java threads.
- Migrated legacy modules to Spring MVC and MySQL behind new REST APIs, reducing development workload by 21%.

PROJECTS

Culdesac | yourculdesac.com 2025 – 2026

- Built and launched a local event-discovery app on Google Play as sole engineer: React Native/Expo client, Supabase Postgres backend, 12 Deno edge functions, 76 migrations, 160+ row-level security policies.
- Wrote feed ranking as SQL RPCs (interests, clubs, distance decay, freshness) with keyset pagination so live-scored rows never drop mid-scroll; replaced full-table scans with trigram and partial geo indexes.
- Automated delivery with 13 GitHub Actions workflows: ∼870 Jest tests per PR, 290 pgTAP assertions on migration changes, native changes routed to EAS builds and JS-only changes shipped as OTA updates.

GroceryPulse | grocerypulse.ca 2026

- Built a Canadian grocery price index and data API: scrapers for 22 retail banners (∼160 stores, 13 cities) load ∼8K prices/week into append-only Postgres with provenance and point-in-time snapshots.
- Replaced headless Chromium with a TLS-fingerprinted HTTP client, cutting per-request latency from 80 s to 0.7 s (110x) with 100% delivery on bot-protected sites.
- Matched 44.6K client SKUs to 31K competitor products (barcodes, brand, pack size, POS-abbreviation expansion, rubric-guided LLM-agent review), raising priced coverage from 33% to 43.5% with 29/30 spot-check precision.

Blind Pear | iOS App Store 2025 – 2026

- Co-built a dating app live on the App Store; made matching race-free with transaction-scoped advisory locks on sorted user pairs and a realtime match queue with atomic claims.
- Traced OTA updates reaching 0 installs to a build variant altering the Expo fingerprint; fixed with a regression test.

Memory 2026

- Built a private, fully on-device AI assistant for iOS/Android (0.6B LLM via llama.cpp, int8 CLIP) with hybrid retrieval (SQLite FTS5 + sqlite-vec + CLIP vectors, reciprocal-rank fusion) over an encrypted store.

E-Ink Sync | GitHub 2026

- Open-source Obsidian plugin (340+ downloads) that syncs reMarkable tablets without the vendor cloud: hand-written parser for legacy binary stroke files, incremental SFTP sync tuned for a 512 MB single-core device, 700+ tests.

EDUCATION

University of Toronto 2019 – 2023
B.Sc. Computer Science, GPA 3.98/4.0

- Teaching Assistant, CSCB58 Computer Organization; Research Intern, embARC Research Group (sparse matrix formats)

SKILLS

Languages: Python, TypeScript/JavaScript, Java, SQL, Hack, C, Elixir, HTML/CSS
Tools: React, React Native/Expo, Next.js, Node.js, PyTorch, PostgreSQL/Supabase, Docker, Kubernetes, GitHub Actions